The wall rings are $\star 2$ for every n (whole-board game)

Model. Whole-board game on C_n (no splitting into components; race rule "count"; e.p. on; a pawn that can still be taken e.p. is not passed; an e.p. flag is recorded only if an enemy pawn stands beside the landing square). A position is *terminal* if exactly one side has a passed pawn, or both do and that side wins the race; in $G + \star h$ ($h < 256$) its holder wins whoever moves. A move μ of X to c (Y the opponent) is T (L) if c is terminal and X wins (loses) it; P (poisoned) if c is not terminal but Y has a move to a terminal position won by Y; K otherwise. The *reduced game* keeps the T-moves (to a sink won by X), the K-moves and the heap moves.

The rings. A file is written by its pawns ($W23B4$: white on ranks 2, 3, black on 4). O_n ($n = 2k+3 \geq 5$): $W23B4$, $B5$, $W2B5$, $(W3, B5)^k$. E_n ($n = 2m+6 \geq 6$): $B5$, $W2B5$, $W35B6$, $W2$, $(B4, W2)^{m+1}$. Both have $n+3$ pawns. Files $3, \ldots, n-1$ form the *wall*: single pawns alternating W3/B5 (type O) or W2/B4 (type E), the black one two ranks above the white one. A *pattern file* is a wall file that still holds its starting pawn.

Theorem 1. *In the whole-board game $O_n = \star 2$ for odd $n \geq 5$ and $E_n = \star 2$ for even $n \geq 6$.*

Lemma 1 (Reduction; [P])**.** *A side wins $G + \star h$ moving first iff it wins the reduced game moving first.*

Proof. The game is finite; a side wins iff some move leads to a position lost by the other side moving first. T-moves win. After an L- or P-move the opponent wins moving first (by the terminal position, resp. the poisoning reply), so these moves never win and may be deleted; K- and heap moves keep their values by induction. □

Lemma 2 (Wall pawns; [P])**.** *Let c have no passed pawn, with an e.p. flag (if any) protecting a pawn of the player not to move, and let q be a wall pawn of the player Z to move, with pattern files $j-1, j, j+1$. (a) After any move of q, Z has no passed pawn. (b) If c has no flag, $Z = X$, and files $j-1, \ldots, j+3$ (or $j-3, \ldots, j+1$) are pattern files, every move of q is L or P.*

Proof. Neighbours of q are pawns of the other colour two ranks further on, so q can only advance; a pawn's status depends on the enemy pawns on its three files and on the flag. (a) The other pawns of Z keep their status (no enemy pawn moved; a lapsing flag protects an enemy pawn). After a step q is still blocked by a neighbour; a white $2 \to 4$ step lands beside a B4 and is protected e.p. (b) By (a) X has no passed pawn afterwards. If Y has one, the move is L. Otherwise Y takes q (e.p. after $2 \to 4$) with his pawn on $j+1$: the capturer stands on j with no X pawn on files $j-1, j, j+1$, so it is passed, and the one X pawn whose status changed, on $j+2$, is still blocked by the Y pawn on $j+3$. So the move is P. □

Lemma 3 (Insertion; [P])**.** *Let p on C_n have no passed pawn and no e.p. flag, and let files $c-4, \ldots, c+5$ be pattern files. Let $\hat{p} = \mathrm{ins}_c(p)$ on C_{n+2} be p with two files, with the contents of files $c+1$ and c, inserted after file c (the pattern continues). For each side, the children of its K-moves in $\hat{p}$ are the ins_c-images of those in p, and it has a T-move in $\hat{p}$ iff in p.*

Proof. $\hat{p}$ has no passed pawn. Every pawn of p has the same three-file view in $\hat{p}$ (files $c, c+1$ carry the pattern): same moves, same status. (i) Pawns on files $c-3, \ldots, c+4$ of p and $c-3, \ldots, c+6$ of $\hat{p}$ (new pawns included) have a window as in Lemma 2(b) inside the pattern arc: all their moves are L or P. (ii) A move μ of a pawn on a file $\leq c-4$ or $\geq c+5$ touches only files $\leq c-3$ or $\geq c+4$: its children are c_μ and $\mathrm{ins}_c(c_\mu)$, with the same passed pawns (new pawns stay blocked), hence the same class T/L/non-terminal (a race compares the same pawns). (iii) If c_μ is not terminal, the replies of Y by pawns on files $\leq c-2$ or $\geq c+3$ correspond, with the same children and, by (ii), the same result; replies by pawns on files $c-1, \ldots, c+2$ and by inserted pawns never win for Y by Lemma 2(a) (files $c-2, \ldots, c+3$ still carry the pattern in c_μ). So μ is P in p iff in $\hat{p}$. By (i)–(iii) the T- and K-moves correspond. □

Finite checks. Let $n_0 = 17$ (type O), 16 (type E).

C1 [C2] The reduced game of O_{n_0} (E_{n_0}) has 144 (104) nodes; no node has an e.p. flag; every node agrees with the start position outside the files $\{n-1, 0, \ldots, 4\}$ ($\{n-1, 0, \ldots, 3\}$); $G + \star 2$ is a P-position. (Two independent programs build the same graph; an unreduced search, and the canonical form of the graph computed with `cylcgt`'s CGT arithmetic, both give the last claim.)

C2 [C2] $G_n + \star 2$ is a P-position for $5 \leq n < n_0$ (unreduced search; whole-board `cylcgt` for $n \leq 10$), and for all $n \leq 61$.

Proof of the Theorem. Let $c = 9$ (type O), 8 (type E). Let $n \geq n_0$ and suppose no node of the reduced game of G_n has a flag and all agree with the start outside the zone of C1 (true for n_0). Then files $c-4, \ldots, c+5$ (= 5..14, resp. 4..13) are pattern files in every node p, and Lemma 3 gives: the successors and T-flags of $\mathrm{ins}_c(p)$ are the ins_c-images of those of p. As $\mathrm{ins}_c(\mathrm{start}_n) = \mathrm{start}_{n+2}$, induction on the distance from the start shows that the reduced game of G_{n+2} is the ins_c-image of that of G_n (nodes, edges, T-flags); its nodes again have no flag and agree with the start outside the shifted zone. So the reduced games of one parity are all isomorphic to the one at n_0; by Lemma 1 and C1 $G_n + \star 2$ is a P-position, i.e. $G_n = \star 2$. For $n < n_0$ use C2. [P] □

Details (not part of the one-column proof)

Reachable positions. By C1 and Lemma 3, for every $n \geq n_0$ the reduced game has exactly 144 (type O) resp. 104 (type E) nodes. A node is the start position with the contents of the *zone* files $n-1, 0, 1, 2, 3, 4$ (O) resp. $n-1, 0, 1, 2, 3$ (E) replaced by one of the listed zone states (files `zone_states_odd.txt`, `zone_states_even.txt`); the zone holds the three window files and the wall pawns next to them (O: B5 on $n-1$, W3 on 3, B5 on 4; E: W2 on $n-1$ and on 3). The wall pawns on files $5, \ldots, n-2$ (O) resp. $4, \ldots, n-2$ (E) never move and are never captured.

The five poisoned wall moves (Lemma 2(b)).

type	move of q on file j	reply of Y	why X has no passed pawn afterwards
O	W3→4	B5 on j+1 takes on $(j,4)$	W3 on j+2 blocked by B5 on j+3
O	B5→4	W3 on j+1 takes on $(j,4)$	B5 on j+2 blocked by W3 on j+3
E	W2→3	B4 on j+1 takes on $(j,3)$	W2 on j+2 blocked by B4 on j+3
E	W2→4	B4 on j+1 takes e.p. on $(j,3)$	same
E	B4→3	W2 on j+1 takes on $(j,3)$	B4 on j+2 blocked by W2 on j+3

In each case the capturer is passed because files $j-1, j, j+1$ then hold no pawn of X.

Remarks. (1) Only the whole-board game is covered. In the component-sum model of `cylcgt`, Lemma 1 is not available: a decisive passer in one component can be cancelled by a $-\mathsf{B}$ option in another. (2) Radius 5 in Lemma 3 is what the hand proof needs; random stress tests find no counterexample for radius ≥ 3 (and find some for radius 1, and for a wrongly phased insertion). (3) The same proof applies to the single-step family F_n (core $W3B6, B6, W4B7, W4, W6B7$ plus a type-E wall): its reduced games have 662 nodes (n even, $n_0 = 18$, cut $c = 4$) and 3568 nodes (n odd, $n_0 = 19$, cut $c = 6$); C1 and C2 were run for them too.